Sample document

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1 LLAlignAnd

`&=`	$a = b$ $c = d$	ok
`=&`	$a =b$ $c =d$	ng
`={}&`	$a = b$ $c = d$	ok

$x =y$	$x \leq y$	$x \leq y$	$x < y$	$x \succ y$	$x \geq y$	$x \approx y$
$x \neq y$	$x \geq y$	$x \geq y$	$x > y$	$x \prec y$	$x \leq y$	$x \asymp y$

verbatim does not care about this rule: `a =& b`

2 LLAlignEnd

The following ends with a line break.

$$f(x) = ax^2 + bx + c \tag{1}$$
$$g(x) = dx^2 + ex + f \tag{2}$$
$$\tag{3}$$

The following does not end with a line break.

$$f(x) = ax^2 + bx + c \tag{4}$$
$$g(x) = dx^2 + ex + f \tag{5}$$

Here is the next line after the align environment.

3 LLAlignSingleLine

Long line before display (same result)

Lorem ipsum.

$$f(x) = ax^2 + bx + c$$

This is an equation environment.

Lorem ipsum.

$$f(x) = ax^2 + bx + c$$

This is an align environment.

Short line before display (different result)

Lrm:
$$f(x) = ax^2 + bx + c$$
This is an equation environment.

Lrm:

$$f(x) = ax^2 + bx + c$$

This is an align environment.

Single-line alignat environment is also detected.

$$f(x) = ax^2 + bx + c$$

Multi-line alignat environment is not detected.

$$\begin{aligned} f(x) &= ax^2 + bx + c \\ g(x) &= dx^2 + ex + f \end{aligned}$$

4 LLArticle

A n-dim. An n-dim. a n-dim. an n-dim. A n dim. An n dim. a n dim. an n dim.

4.1 Single Letters

OK: an a-dim, a b-dim. a u-dim.

- NG: ‘a a‘ → OK: ‘an a‘
- NG: ‘a e‘ → OK: ‘an e‘
- NG: ‘a f‘ → OK: ‘an f‘
- NG: ‘a h‘ → OK: ‘an h‘
- NG: ‘a i‘ → OK: ‘an i‘
- NG: ‘a l‘ → OK: ‘an l‘
- NG: ‘a m‘ → OK: ‘an m‘
- NG: ‘a n‘ → OK: ‘an n‘

- NG: ‘a o‘ → OK: ‘an o‘
- NG: ‘a r‘ → OK: ‘an r‘
- NG: ‘a s‘ → OK: ‘an s‘
- NG: ‘a x‘ → OK: ‘an x‘
- NG: ‘an u‘ → OK: ‘a u‘

4.2 Abbreviations

- NG: ‘a EM‘ → OK: ‘an EM‘ (Expectation–Maximization)
- NG: ‘a EVD‘ → OK: ‘an EVD‘ (Eigenvalue Decomposition)
- NG: ‘a FFT‘ → OK: ‘an FFT‘ (Fast Fourier Transform)
- NG: ‘a NP-hard‘ → OK: ‘an NP-hard‘ (Non-deterministic Polynomial-time hard)
- NG: ‘a LSTM‘ → OK: ‘an LSTM‘ (Long Short-Term Memory)
- NG: ‘a LTI‘ → OK: ‘an LTI‘ (Linear Time-Invariant)
- NG: ‘a MLE‘ → OK: ‘an MLE‘ (Maximum Likelihood Estimation)
- NG: ‘a MSE‘ → OK: ‘an MSE‘ (Mean Squared Error)
- NG: ‘a ODE‘ → OK: ‘an ODE‘ (Ordinary Differential Equation)
- NG: ‘a RNN‘ → OK: ‘an RNN‘ (Recurrent Neural Network)
- NG: ‘a RKHS‘ → OK: ‘an RKHS‘ (Reproducing Kernel Hilbert Space)
- NG: ‘a SDE‘ → OK: ‘an SDE‘ (Stochastic Differential Equation)
- NG: ‘a SVD‘ → OK: ‘an SVD‘ (Singular Value Decomposition)
- NG: ‘a SVM‘ → OK: ‘an SVM‘ (Support Vector Machine)
- NG: ‘a XOR‘ → OK: ‘an XOR‘ (Exclusive OR)

4.3 LaTeX Commands

- NG: ‘a $\mathbb{R}$-valued function‘ → OK: ‘an $\mathbb{R}$-valued function‘
- NG: ‘a L^1 norm‘ → OK: ‘an L^1 norm‘
- NG: ‘a L^2 norm‘ → OK: ‘an L^2 norm‘
- NG: ‘a L^∞ norm‘ → OK: ‘an L^∞ norm‘
- NG: ‘a ℓ^1 norm‘ → OK: ‘an ℓ^1 norm‘
- NG: ‘a ℓ^2 norm‘ → OK: ‘an ℓ^2 norm‘
- NG: ‘a ℓ^∞ norm‘ → OK: ‘an ℓ^∞ norm‘
- NG: ‘a $\mathcal{L}^1$ norm‘ → OK: ‘an $\mathcal{L}^1$ norm‘
- NG: ‘a $\mathcal{L}^2$ norm‘ → OK: ‘an $\mathcal{L}^2$ norm‘
- NG: ‘a $\mathcal{L}^\infty$ norm‘ → OK: ‘an $\mathcal{L}^\infty$ norm‘

4.4 Not Detected

An $\mathbb{R}$-valued, a *abced*-dim.

5 LLBig

This is a sample text. This is a sample text. This is a sample text.
Both bigcup $\bigcup_{x\in B} O_x$ and cup $\cup_{x\in B} O_x$ do not spoil the line spacing.
This is a sample text. This is a sample text. This is a sample text.

$$X_1 \cap X_2 \quad X_1 \cup X_2 \quad X_1 \odot X_2 \quad X_1 \oplus X_2 \quad X_1 \otimes X_2$$
$$X_1 \sqcup X_2 \quad X_1 \uplus X_2 \quad X_1 \vee X_2 \quad X_1 \wedge X_2 \quad \text{ok}$$
$$\bigcap_{i=1}^{\infty} X_i \quad \bigcup_{i=1}^{\infty} X_i \quad \bigodot_{i=1}^{\infty} X_i \quad \bigoplus_{i=1}^{\infty} X_i \quad \bigotimes_{i=1}^{\infty} X_i \quad \bigsqcup_{i=1}^{\infty} X_i \quad \biguplus_{i=1}^{\infty} X_i \quad \bigvee_{i=1}^{\infty} X_i \quad \bigwedge_{i=1}^{\infty} X_i \quad \text{ok}$$
$$\cap_{i=1}^{\infty} X_i \quad \cup_{i=1}^{\infty} X_i \quad \odot_{i=1}^{\infty} X_i \quad \oplus_{i=1}^{\infty} X_i \quad \otimes_{i=1}^{\infty} X_i$$
$$\sqcup_{i=1}^{\infty} X_i \quad \uplus_{i=1}^{\infty} X_i \quad \vee_{i=1}^{\infty} X_i \quad \wedge_{i=1}^{\infty} X_i \quad \text{ng}$$

6 LLBracketCurly

`\max(a,b)`	$\max(a,b)$	ok
`\max{a,b}`	$\max a,b$	ng
`\max {a,b}`	$\max a,b$	ok?

We cannot fully determine whether the use of curly brackets is wrong or not. It is not detected if some spaces are inserted between the command name and the curly brackets.
$\min(a,b)$ and $\min a,b$ are also checked.

7 LLBracketMissing

`x^{23}`	x^{23}	ok
`x^2 3`	$x^2 3$	ok
`x^23`	$x^2 3$	ng

$x_2 3$, $x^a b$, and $x_a b$ are also checked. Cases like $x^a b$, x^2 , and $e^i \pi$ are not detected. Disabled by default. (It contains false positives, and may be too strict for some users.)
Following examples are ignored.

```
escaped underscore: \_123
https://sample_url.com
\includegraphics{sample_link.png}
\includegraphics{sample_link.pdf}
\label{sample_label}
\cref{sample_label}
\eqref{sample_label}
\cite{sample_label}
```

```
\Cref{sample_Label}
\bibliographystyle{my_style}
\addbibresource{ref_file.bib}
% x^23 comment
```

8 LLBracketRound

`\sqrt{a}`	$\sqrt{a}$	ok
`\sqrt(a)`	$\sqrt(a)$	ng

$a^(1)$ and $a_(1)$ are also checked.
Following examples are ignored:
`Test\_(1)` Test_(1)
`Test\^2` Test$\hat{2}$
`sample_(1).png` sample_(1).pdf

Figure 1: sample_(1).png

`\label{eq:f(x_(k+1)) <= m(x_(k+1)) <= m(x_k) = f(x_k)}`

$$f(x_{k+1}) \leq m(x_{k+1}) \leq m(x_k) = f(x_k) \tag{6}$$

The label in Eq. (6) is ignored.

9 LLColonEqq

`\coloneqq`	$x \coloneqq y$	ok
`\Coloneqq`	$x \Coloneqq y$	ok
`:=`	$x := y$	ng
`::=`	$x ::= y$	ng

The difference is quite subtle, but the vertical position of the colon is different. This difference become clearer with some other math fonts.

10 LLColonForMapping

`A:B`	$A : B$	ok
`A\colon B`	$A\colon B$	ok?
`f:`	$f : \mathbb{R} \to \mathbb{R}$	ng
`f\colon`	$f\colon \mathbb{R} \to \mathbb{R}$	ok

We detect all of : in the following

Here are examples of colons we detect.

- $f : X \to Y$
- $g : X \mapsto Y$
- $h : \mathbb{R}^{n^2+2n+1} \to \mathbb{R}$

and

$$f : (X \text{ at new line in tex file}) \to (Y \text{ at new line in tex file}). \tag{7}$$

We do NOT detect any of : in the following

Here are examples of ‘:’ we do not detect.

- $f\colon X \to Y$, the correct use of \colon.
- $A : B : C = 1 : 2 : 3$, the colon for ratio.
- $A : B = 1 : 2$ and $X \to Y$, separated by dollar sign.
- $g :$ (some very very very very very long long long long words) $\to \mathbb{R}$, the false negative.

The following contains colon: $f : X \to Y$
The following contains colon: $f : X \to Y$
The following contains colon:

$$f : X \to Y \tag{8}$$
$$\lim_{k \to \infty} x_k = L \tag{9}$$

11 LLCref

Theorem 1. *This is a sample theorem.*

Use Thm. 1 with cref instead of Thm. 1 with ref to avoid mistakes. Disabled by default.

Algorithm 1 Sample Algorithm

```
1: First line
2: Second line
```

In the pseudocode, cleveref can fail to appropriately reference lines. Thus, we don’t detect `\ref{line:1}` like this: Line 1. We detect other formats `\ref{myline:2}` like this: Line 2. You can change these behaviors via settings.

12 LLDoubleQuotes

Use “XXX” instead of “XXX” or ”XXX”. You can use them for `H\"older` and `\verb`.
`"Checking LLText..."`

```
"double quotes in verbatim"
```

```
def test():
        S = "double quotes in listing"
```

13 LLENDash

`hyphen (-)`	A-B
`en-dash (--)`	A–B
`em-dash (---)`	A—B

Examples:

- Erdos-Renyi (random graph, Erdős–Rényi)
- Einstein-Podolsky-Rosen (quantum physics, Einstein–Podolsky–Rosen)
- Fruchterman-Reingold (graph drawing, Fruchterman–Reingold)
- Gauss-Legendre (numerical integration, Gauss–Legendre)
- Gibbs-Helmholtz (thermodynamics, Gibbs–Helmholtz)
- Karush-Kuhn-Tucker (optimization, Karush–Kuhn–Tucker)

Exceptions: Non-Negative, Well-Known, etc.

14 LLEqnarray

We should not use eqnarray. It has some spacing issues.

$$\begin{array}{rcl} x & = & y \\ a & = & b \end{array}$$

15 LLErrCompOps

$$x \leq y \quad x \geq y \quad x = y \quad \text{ok}$$

$$x <= y \quad x <= y \quad x =< y \quad x =< y \quad x \leq= y \quad x =\leq y \quad x \leq= y \quad x =\leq y \quad \text{ng}$$

$$x => y \quad x => y \quad x >= y \quad x >= y \quad x \geq= y \quad x =\geq y \quad x \geq= y \quad x =\geq y \quad \text{ng}$$

16 LLFootnote

16.1 Bad Examples

Space before [1]
Tab before [2]
Line break before [3]
Line break before with comment [4]
Space before percent sign [5]

16.2 Good Examples

Correct attach.[6]
Phrase attach[7] ok.

Line break suppressed[8] Line break suppressed[9]

Inserted comment[10]
Emphasis *text*[11].

17 LLHeading

17.0.1 Incorrect Hierarchy

17.1 First Subsection

17.1.1 Second Subsubsection

18 LLLlGg

`\ll`	$n \ll m$	ok
`<<`	$n << m$	ng

I like human <<< cat <<<<<<< dog.

---

[1] bad footnote
[2] bad footnote
[3] bad footnote
[4] bad footnote
[5] bad footnote
[6] good footnote
[7] good footnote
[8] good footnote
[9] good footnote
[10] good footnote
[11] good footnote

19 LLNonASCII

The following line contains non-ASCII characters.
　！＂＃＄％＆＇（）＊＋－／

日本語の文章は、upLaTeXでフツウに書けます。
(You can write Japanese sentences as usual with upLaTeX.)

20 LLNonstandard

20.1 Therefore and Because

Not Standard?: $\therefore a = b$, $\because x > 0$
Standard: Therefore, $a = b$. Because $x > 0$, ...

20.2 Iff

Not Standard?: This statement is true iff the condition holds.
Standard: This statement is true if and only if the condition holds.
OK: `\iff` is a valid LaTeX command and should not be detected. $a = b \iff c = d$.

20.3 DotSeq

Not Standard?: $a \fallingdotseq b$, $x \risingdotseq y$
Standard: $a \approx b$, $x \approx y$

20.4 Combination Notation

Not Standard?: ${}_nC_k$, ${}_n\mathrm{C}_k$
Standard: $\binom{n}{k}$

21 LLPeriod

The period character (.) can be used in abbreviations, not as a sentence-ending period. In the former case, it should be followed by a backslash to prevent LaTeX from treating it as a sentence-ending period, which can affect spacing.

`e.g., Test.`	e.g., Test.	ok
`e.g.\ Test.`	e.g. Test.	ok
`e.g. Test.`	e.g. Test.	ng

Without a backslash: e.g. blah i.e. blah i.i.d. blah w.r.t. blah w.l.o.g. blah resp. blah etc.
With a backslash: e.g. blah i.e. blah i.i.d. blah w.r.t. blah w.l.o.g. blah resp. blah etc.

22 LLRefEq

To refer to the equation, use (7) with eqref instead of (7) with ref.
Eq. 7 and Prob. 7 are allowed.

23 LLSharp

Some people mistake `\sharp` to be a symbol for the number of elements in a set, but it is not. It is a musical symbol. The correct symbol for the number of elements in a set is `\#`.

`\#`	$\#A$	ok
`\sharp`	$\sharp A$	ng

- matching examples: $\sharp A$, $\sharp\{1,2,3\}$, $\sharp X_0$
- non-matching examples: $\sharp$, $f^\sharp X$, $\alpha_\sharp X$, $C\sharp$

If you really want to use $\sharp$, you can disable this rule.

24 LLSI

`\SI{1}{\kilo\byte}`	1 kB	ok
`1 kB`	1 kB	ng
`1kB`	1kB	ng

10KB, 3.5 MiB, 500GB are detected. 123 noNumWord GB will not be detected.
Some command named as EB. This is not ExaByte. This 1EB is one ExaByte.

25 LLSortedCites

Unsorted citations [2, 1]: ng?
Automatically sortable like this: [1, 2].

26 LLSpaceEnglish

Missing spaces around inline math should be detected.
Improper spacing: xyz and xyz. Proper spacing: x y z and x y z.
The following are excluded by rule exceptions: nth, mth, \n, \m, H_2O, C_nH_{2n+2}.
No detection in the following cases: 1x, 1x, x1, x1, x, x, etc.

27 LLSpaceJapanese

The following are Japanese examples, which are excluded by rule exceptions.
日本語の文章で$x = 1$もしくは$y = 2$と数式を書くと、スペースが欠如します。
日本語の文章で $x = 1$ もしくは $y = 2$ と数式を書くと、スペースが生まれます。
尤も、フォーマルな文章では非推奨な場合もあり、その為デフォルトでは無効です。
テスト:これは大丈夫。テスト(これは大丈夫)。1文字だけ間にある場合としてxとyのテスト。

28 LLT

`^\top`	$X^\top$	ok
`^{\mathsf{T}}`	X^{T}	ok
`^T`	X^T	ng
`^{T}`	X^{T}	ok?

Disabled by default. The following cases are not detected as errors.

$$\sum_{i=1}^{T} x_i, \quad \sum_{0}^{T} y_i, \quad \sum_{\alpha}^{T} z_i, \quad \sum_{i=1}^{T} y_i,$$
$$\prod_{i=1}^{T} x_i, \quad \prod_{0}^{T} y_i, \quad \prod_{\beta}^{T} z_i, \quad \prod_{i=1}^{T} y_i,$$
$$\int_0^T f(t)dt, \quad \int_0^T f(t)dt, \quad \int_\gamma^T f(t)dt$$

29 LLTextLint

Only for Japanese texts.
ご静聴ありがとうございました。 御静聴ありがとうございました。 まず第1に、背景を説明します。 まず第一に、背景を説明します。 このモデルをシュミレーションします。 この命題は帰納法で証明できます。 1番最初に確認します。 一番最初に確認します。 一番最後に結論を述べます。 この実装はまだ未完成です。 開催日時はまだ未定です。 この問題はまだ未解決です。 これは第1回目です。 今日は第2日目です。 この関数の2回導関数を考えます。
Sample text: Hamaguchi et. al. provide supplemental materials on Youtube, Github and VScode Marketplace, which contains Latex source code. This sentence contains errors we will detect by LLTextLint.

30 LLThousands

`$12, 34$`	$12, 34$	ok
`1,000`	1,000	ok
`$1{,}000$`	$1{,}000$	ok
`$1,000$`	$1,000$	ng

31 LLTitle

31.1 This Is a Correct Title

31.1.1 this is a wrong title

The quick brown fox jumps over the lazy dog

SubParagraph: Test With Ref 1

31.2 IGNORE IF ALL UPPERCASE

31.3 Math Contains version x

32 LLUnRef

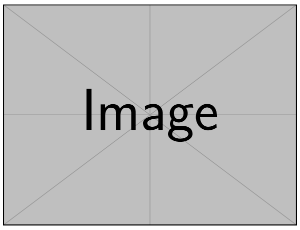

Figure 2: Referenced Figure

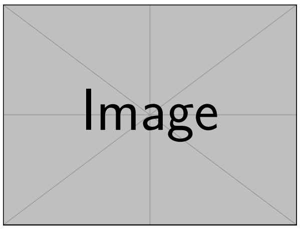

Figure 3: Unreferenced Figure

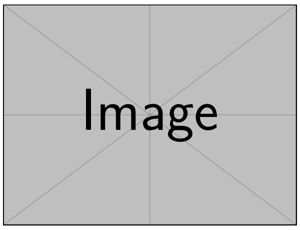

Figure 4: Referenced Figure 2

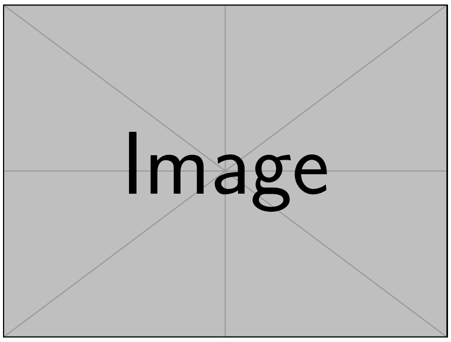

Figure 5: Unreferenced Wide Figure

A	B
1	2

Table 1: Referenced Wide Table

Figure 2, fig. 4, and 1 are referenced in the text, while the other figure/table labels are not.

33 LLURL

The following URLs contain unnecessary query strings (should be detected):

- `https://example.com/page?utm_source=newsletter&utm_medium=email`
- `https://example.com/page?sessionid=123456`

The following URLs contain only allowed query strings (should not be detected):

- `https://example.com/search?q=latex`

C	D
3	4

Table 2: Unreferenced Wide Table

- `https://example.com/list?page=2`
- `https://example.com/view?lang=ja`

Examples of URLs without query strings:

- `https://example.com/clean-page`

34 LLUserDefined

You can define your own rule.

`f^\mathrm{a}(x)`	$f^{\mathrm{a}}(x)$	ok
`f^a(x)`	$f^a(x)$	ng

`f \infConv g`	$f \Box g$	ok
`f \Box g`	$f \Box g$	ng

References

[1] A. Article. *Journal*, 2000.

[2] B. *Book*. Publisher, 2000.

Appendix A Check LLText

```
Wrong LaTeX math environment:
\begin{align}
\end{equation}
```

Appendix B LLSetBar

Detecting inappropriate use of the vertical bar | is very difficult. We are currently trying to detect the following, although not implemented yet.

`\lvert -1 \rvert`	$\lvert -1 \rvert$	ok
`\abs{-1}`	$\lvert -1 \rvert$	ok
`\vert -1 \vert`	$\vert -1 \vert$	ng
`|-1|`	$\vert -1 \vert$	ng

`\lVert -x \rVert`	$\lVert -x \rVert$	ok
`\norm{-x}`	$\lVert -x \rVert$	ok
`\Vert -x \Vert`	$\Vert -x \Vert$	ng
`||-x||`	$\|\| -x \|\|$	ng

`\relmiddle|` (macro)	$\left\{ a \;\middle|\; a > \frac{1}{2} \right\}$	ok
`\mid`	$\{ a \mid a > \frac{1}{2} \}$	ok?
`|`	$\{ a \vert a > \frac{1}{2} \}$	ng

`\divides` (MnSymbol)	$+2 \mid +4$	ok
`\mid`	$+2 \mid +4$	ok?
`\mathrel|`	$+2 \mathrel{|} +4$	ok?
`\vert`	$+2 \vert + 4$	ng
`|`	$+2 \vert + 4$	ng

`f(y|x)`	$f(y \vert x)$	ok?
`f(y \mid x)`	$f(y \mid x)$	ok?
`f(\,y\mid x\,)`	$f(\,y \mid x\,)$	ok?
`\left. \dv{t} \right|_{t=0} f(t)`	$\left. \frac{\mathrm{d}}{\mathrm{d}t} \right\vert_{t=0} f(t)$	ok?